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Thermodynamic Optimization of the Sc-X (X = La, Li, Ca, Cr) Systems

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The Sc-X (X = La, Li, Ca, Cr) systems have been thermodynamically assessed by means of the CALPHAD (CALculation of PHAse Diagram) method. For each of the binary systems, a set of self-consistent thermodynamic parameters has been obtained. Comparisons between the calculated and measured phase diagrams show that the experimental data are satisfactorily accounted for by the present thermodynamic modeling.

Keywords CALPHAD, phase diagram, Sc-Ca system, Sc-Cr system, Sc-La system, Sc-Li system, thermodynamic modeling

1. Introduction

Sc, La, Li, Ca and Cr are common additive elements of aluminum alloys. The rare-earth elements Sc and La can remove gases and harmful impurities and reduce the occurrence of cracks during solidification in aluminum alloys. With the addition of a small amount of Sc or La, the tensile strength, elongation and hardness, as well as the corrosion and heat resistance of aluminum alloys can be significantly improved.[1] The alkali metals Li and Ca have a negative effect on the strength of aluminum alloys as impurity elements. However Li can reduce the density and enhance the specific strength of aluminum alloys.[2,3] In addition, the alloying element Cr can improve the toughness, and increases the resistance to stress corrosion cracking of aluminum alloys.[4] Currently, the establishment of a thermodynamic database for multi-component Al alloys is in progress in our group.[5-10] The binary systems Sc-X (X = La, Li, Ca and Cr) are the subsystems of the multi-component Al-based alloys and their thermodynamic descriptions are the prerequisite for the development of the thermodynamic database.

Up to now, no thermodynamic modeling has been performed for the Sc-X (X = La, Li, Ca, Cr) systems. The main purpose of the present work is to obtain a set of self-consistent thermodynamic parameters for each of these binary subsystems by means of the CALPHAD approach.

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2. Literature Review

2.1 Phase Diagram Data

No intermetallic compound has been reported in these four binary systems.

The phase diagram of the Sc-La system was studied by Naumkin et al.,[11] using microstructural analysis, x-ray diffraction and thermal analysis. Two eutectoid reactions, (γLa, βSc) ⇌ (βLa) + (αSc) and (βLa) ⇌ (αLa) + (αSc), were detected to occur at 750 ± 5 °C, ~45 at.% Sc, and to occur at 233 ± 2 °C, ~14 at.% Sc, respectively. The solubility of Sc in (βLa) at 750 °C and in (αLa) at 233 °C was determined by microstructural analysis to be ~16 at.% Sc and ~12 at.% Sc, respectively. The Sc-La system was reviewed by Gschneidner and Calderwood.[12] The early measured allotropic transformation temperature (310 °C) of La by Spedding et al.[13] was adopted by Naumkin et al.[11] and by Gschneidner and Calderwood.[12] However, the currently accepted temperature for the allotropic transformation in the SGTE compilation[14] is 277 °C, which is 33 °C lower than the measured temperature by Spedding et al.[13] In the present work, we adopt the thermodynamic data from SGTE, since Spedding et al.[13] used the method of dilatometric measurements and found that the transition temperature of Lanthanum on cooling is quite low and the transition appears to be quite sluggish, so they only considered the heating curves, and this cannot be considered accurate.

The Sc-Li phase diagram was first measured by Ganiev et al.[15] and later reviewed by Okamoto.[16] There is a miscibility gap for the liquid phase. Two binary invariant reactions were detected in this system. One is Liquid#2 = Liquid#1 + (αSc) occurring at 970 °C and ~58.3 at.% Sc for the Liquid#2 and another one is Liquid#1 + (αSc) = (βLi) occurring at 180 °C and ~0.07 at.% Sc for the Liquid#1.

The Sc-Ca phase diagram is mainly based on the work of Moffatt.[17] The temperature for the monotectic reaction Liquid#2 ⇌ Liquid#1 + (βSc) was detected to be 1462 °C at about 95.4 at.% Sc. The maximum solubility of Ca in (βSc) was measured to be 2.6 at.% Sc at this temperature.

The Sc-Cr phase diagram was established by Svechnikov et al.[18] using differential thermal analysis

(DTA), microhardness determinations, x-ray diffraction and metallographic examination and was later reviewed by Venkatraman and Neumann.[19] The eutectoid reaction temperature for (βSc) ⇌ (βCr) + (αSc) and the eutectic reaction temperature for Liquid ⇌ (βCr) + (βSc) were determined by Svechnikov et al.[18] to be 990 °C at 60-65 at.% Sc and to be 1090 °C, respectively. According to Svechnikov et al.,[18] the solubility of Cr in (βSc) should be less than 10 at.% Cr and the maximum solubility of Sc in (βCr) is 2.0 at.% at 1090 °C. However, it should be mentioned that the solubility limit of Sc in (βCr) was measured to be less than 0.35 at.% Sc by Geiselman[20] using metallography, and to be less than 0.1 at.% Sc by Taylor[21] using x-ray diffraction. The result of Taylor[21] is considered to be more reliable and utilized in the present modeling.

2.2 Thermodynamic Data

No experimental thermodynamic data on these binary systems are available in the literature. In the present work, the highly-efficient first-principles plane wave pseudopotential method as implemented in the Vienna ab initio simulation package (VASP)[22,23] is utilized for the present first-principles calculations to compute the lattice stability of the element La. The exchange-correlation is treated within the Generalized Gradient Approximation (GGA) of Perdew-Burke-Ernzerhof (PBE).[24] The Monkhorst-Pack scheme[25] k-points sampling together with the linear tetrahedron method including the Blöchl corrections[26] is adopted for the integration in the Brillouin zone (BZ).

The lattice stability of the element La is evaluated by the following equation:

$$E_{\mathrm{La}}^{\mathrm{fcc-hcp}} = E_{\mathrm{La}}^{\mathrm{fcc}} - E_{\mathrm{La}}^{\mathrm{hcp}} \qquad \text{(Eq 1)}$$

where $E_{\mathrm{La}}^{\mathrm{fcc-hcp}}$ is the lattice stability, $E_{\mathrm{La}}^{\mathrm{fcc}}$ and $E_{\mathrm{La}}^{\mathrm{hcp}}$ are the energies (per atom) of the Fcc_A1 and Hcp_A3 configurations of La respectively. In present work ${}^0G_{\mathrm{La:Va}}^{\mathrm{Hcp_A3}}$ was calculated to be ${}^0G_{\mathrm{La}}^{\mathrm{HSER}}$ + 2756.9.

3. Thermodynamic Models

3.1 Unary Phase

The Gibbs energy function ${}^0G_i^{\phi}(T) = G_i^{\phi}(T) - H_i^{\mathrm{SER}}$ for the element i (i = Sc, La, Li, Ca, Cr) in the phase Φ is expressed as follows:

$$\begin{aligned} {}^0G_i^{\phi}(T) = {} & a + b \cdot T + c \cdot T \cdot \ln T + d \cdot T^2 + e \cdot T^{-1} \\ & + f \cdot T^3 + g \cdot T^7 + h \cdot T^{-9} \end{aligned} \qquad \text{(Eq2)}$$

where H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state. The last two terms in Eq 2 are used only outside the ranges of stability,[27] $g{\cdot}T^7$ for a liquid below the melting point and $h{\cdot}T^{-9}$ for solid phases above the melting point. These Gibbs energy functions for Sc, La, Li, Ca and Cr are taken from the SGTE compilation by Dinsdale.[14]

3.2 Solution Phases

The liquid, Fcc_A1 ((βLa) and (αCa)), Bcc_A2 ((βSc), (γLa), (βLi), (βCa) and (βCr)), Hcp_A3 (αSc), Dhcp_A3 (αLa) phases are modeled as completely disordered solutions, and its excess Gibbs energy is described by the Redlich-Kister (R-K) polynomial.[28] The Gibbs energy of the above phases is expressed by the following equation:

$$\begin{aligned} G_{\mathrm{m}}^{\phi} - H^{\mathrm{SER}} = {} & x_{\mathrm{Sc}} \cdot {}^0G_{\mathrm{Sc}}^{\phi} + x_i \cdot {}^0G_i^{\phi} \\ & + R \cdot T \cdot [x_{\mathrm{Sc}} \cdot \ln(x_{\mathrm{Sc}}) + x_i \cdot \ln(x_i)] \\ & + x_{\mathrm{Sc}} \cdot x_i \cdot \sum_{j=0}^{n} L_{\mathrm{Sc},i}^{j,\phi}(x_{\mathrm{Sc}} - x_i)^j \end{aligned} \qquad \text{(Eq3)}$$

where H^{SER} is the abbreviation of $x_{\mathrm{Sc}} \cdot H_{\mathrm{Sc}}^{\mathrm{SER}} + x_i \cdot H_i^{\mathrm{SER}}$, R is the gas constant, x_{Sc} and x_i are the mole fractions of Sc and i (i = La, Li, Ca, Cr), respectively. $x_{\mathrm{Sc}} \cdot {}^0G_{\mathrm{Sc}}^{\phi} + x_i \cdot {}^0G_i^{\phi}$ denotes the mechanical mixing of the pure elements. $R \cdot T \cdot [x_{\mathrm{Sc}} \cdot \ln(x_{\mathrm{Sc}}) + x_i \cdot \ln(x_i)]$ corresponds to the contribution of the ideal entropy of mixing to the Gibbs energy. $x_{\mathrm{Sc}} \cdot x_i \cdot \sum_{j=0}^{n} L_{\mathrm{Sc},i}^{j,\phi}(x_{\mathrm{Sc}} - x_i)^j$ denotes the excess Gibbs energy. $L_{\mathrm{Sc},i}^{j,\phi}$ is the jth interaction parameter of solution phase Φ and it is usually expressed as the following, where coefficients $a_j$and b_j (j = 0, 1, 2) are the parameters to be optimized:

$$L_{\mathrm{Sc},i}^{j,\phi} = a_j + b_j \cdot T \qquad \text{(Eq 4)}$$

4. Results and Discussion

The model parameters were evaluated using the PARROT module in Thermo-Calc,[29] which works by minimizing the square sum of the differences between experimental values and computed ones. Each piece of experimental information was given a certain weight in the assessment procedure.

At the beginning of the optimizing procedure, each phase (Liquid, Fcc_A1, Bcc_A2, Hcp_A3 and Dhcp_A3) was described as the regular solution with the interaction parameter ${}^0L_{Sc,i}$ in Eq 3. Only when ${}^0L_{\mathrm{Sc},i}$ is not enough to fit experimental values, higher order interaction parameters ${}^1L_{\mathrm{Sc},i}$ and ${}^2L_{\mathrm{Sc},i}$ were considered. The invariant reaction data were used to describe the general features of phase diagram, and the experimental solubility and liquidus were utilized to refine the preliminary optimization. The step-by-step optimization procedure carefully described by Du et al.[30] was employed in the present work. Taking the Sc-La system as an example, the optimization began with the Liquidus, only considering the Liquid and Bcc_A2 phases. Then the experimental data of the reaction (βLa) ⇌ (αLa) + (αSc) were used and the parameters of Fcc_A1 and Hcp_A3 were utilized in the optimized procedure. Finally, all of the selected experimental data were considered in order to get a set of self-consistent thermodynamic parameters.

In the present work, there is a lack of experimental data except for the Sc-La system, thus the thermodynamic model should not be too complex. However, due to the miscibility

gap of Liquid or Bcc phases, the 1L parameter was utilized in Sc-Li, Sc-Ca and Sc-Cr systems. Especially in the Sc-Li system, the 2L parameter was used to describe the liquid phase. This was necessary to limit the maximum of liquid-liquid miscibility gap at a temperature which is lower than the La boiling temperature (1327 °C).

The presently obtained parameters for the Sc-X (X = La, Li, Ca, Cr) systems are listed in Table 1. The calculated invariant reactions in the Sc-X (X = La, Li, Ca, Cr) systems along with the literature data are listed in Table 2.

Table 1 Summary of the thermodynamic parameters in the Sc-X (X = La, Li, Ca, Cr) systems

System	Phase	Sublattice model	Evaluated parameters
Sc-La	Liquid	$(Sc, La)_1$	${}^0L^{Liquid} = 15704 - 7.3T$
			${}^1L^{Liquid} = 5187$
	Fcc_A1	$(Sc, La)_1Va_1$	${}^0L^{Fcc_A1} = -5174 + 17.8T$
	Bcc_A2	$(Sc, La)_1$	${}^0L^{Bcc_A2} = 15948$
			${}^1L^{Bcc_A2} = 1861$
	Hcp_A3	$(Sc, La)_1Va_{0.5}$	${}^0G^{Hcp_A3}_{La:Va} = {}^0G^{HSER}_{La} + 2756.9$
			${}^0L^{Hcp_A3} = 2907 + 16.6T$
			${}^1L^{Hcp_A3} = 9542$
	Dhcp_A3	$(Sc, La)_1$	${}^0L^{Dhcp_A3} = 1485 + 19.0T$
			${}^1L^{Dhcp_A3} = -3800$
Sc-Li	Liquid	$(Sc, Li)_1$	${}^0L^{Liquid} = 15519$
			${}^1L^{Liquid} = 7913$
			${}^2L^{Liquid} = -5283$
	Bcc_A2	$(Sc, Li)_1Va_3$	${}^0L^{Bcc_A2} = 47776$
			${}^1L^{Bcc_A2} = -14574$
	Hcp_A3	$(Sc, Li)_1Va_{0.5}$	${}^0L^{Hcp_A3} = 58238$
Sc-Ca	Liquid	$(Sc, Ca)_1$	${}^0L^{Liquid} = 48386$
	Fcc_A1	$(Sc, Ca)_1Va_1$	${}^0L^{Fcc_A1} = 56572$
	Bcc_A2	$(Sc, Ca)_1Va_3$	${}^0L^{Bcc_A2} = 68940$
	Hcp_A3	$(Sc, Ca)_1Va_{0.5}$	${}^0L^{Hcp_A3} = 65910$
Sc-Cr	Liquid	$(Sc, Cr)_1$	${}^0L^{Liquid} = -9257 + 11.5T$
			${}^1L^{Liquid} = -21627 + 17.3T$
	Bcc_A2	$(Sc, Cr)_1Va_3$	${}^0L^{Bcc_A2} = 37578 + 1.2T$
			${}^1L^{Bcc_A2} = 10802$
	Hcp_A3	$(Sc, Cr)_1Va_{0.5}$	${}^0L^{Hcp_A3} = 43195$

Table 2 Calculated invariant equilibria in the Sc-X (X = La, Li, Ca, Cr) systems compared with literature data

System	Reactions	T_{exp}, °C	T_{cal}, °C	Reference
Sc-La	(γLa, βSc) ⇌ (βLa) + (αSc)	750	751	[11]
	(βLa) ⇌ (αLa) + (αSc)	233	232	
Sc-Li	(βSc) + Liquid#2 ⇌ (αSc)	1335	1335	[15]
	Liquid#2 ⇌ Liquid#1 + (αSc)	970	970	
	Liquid#1 + (αSc) ⇌ (βLi)	181	180	
Sc-Ca	Liquid#2 ⇌ Liquid#1 + (βSc)	1462	1478	[17]
	(βSc) + Liquid#2 ⇌ (αSc)	1335	1335	
	Liquid#2 ⇌ (βCa) + (αCa)	842	839	
	(βCa) ⇌ (αCa)	443	443	
Sc-Cr	Liquid ⇌ (βCr) + (βSc)	1090	1093	[18]
	(βSc) ⇌ (βCr) + (αSc)	990	993	

Figure 1 shows the calculated phase diagram of the Sc-La system. According to the present calculation, the invariant reaction (γLa, βSc) ⇌ (βLa) + (αSc) occurs at 751 °C and 45.2 at.% Sc. The calculated solubility of Sc in (βLa) and La in (αSc) at 751 °C is 15.7 and 16.4 at.%, respectively. The calculated invariant reaction (βLa) ⇌ (αLa) + (αSc) is located at 232 °C and 12.4 at.% Sc for (βLa). The calculated solubility of Sc in (αLa) and of La in (αSc) at 232 °C is 10.8 and 13.5 at.%, respectively. Generally, the calculated Sc-La phase diagram agrees well with the experimental data.[11] It is worth mentioning that the calculated solidus of (αLa) does not significantly drop with its extension into the binary system due to our adopting the STGE value[14] rather than the experimental value reported by Spedding et al.[13] for the allotropic transformation temperature between (βLa) and (αLa).

Figure 2 presents the calculated phase diagram of the Sc-Li system. According to the calculation, the temperature and composition of the monotectic reaction Liquid#2 ⇌ Liquid#1 + (αSc) are 970 °C and 58.3 at.% Sc for Liquid#2, respectively, which are the same as the experimental ones.[15] The calculated temperature of the reaction Liquid#1 + (αSc) ⇌ (βLi) is 180 °C, which is very close to the experimental value 181 °C[15] besides the same calculated composition of 0.07 at.% Sc. Comparisons between the calculated and measured phase diagrams showed that the measured data are satisfactorily accounted for by the present thermodynamic description.

The calculated phase diagram of the Sc-Ca system is presented in Fig. 3. The assessed phase diagram by Moffatt[17] can be well reproduced by the present calculation. The temperature of the monotectic reaction Liquid#2 ⇌ Liquid#1 + (βSc) was calculated to be 1478 °C at 95.3 at.% Sc, which values are close to the experimental values 1462 °C and 95.4 at.% Sc.[17] The solubility of Ca in (βSc) was calculated to be 1 at.% Sc at this temperature, which is less than half of the experimental result 2.6 at.%. Generally, the presently obtained parameters reproduce well the Sc-Ca phase diagram.

Figure 4 shows the calculated phase diagram of the Sc-Cr system. The calculated phase diagram is in a good agreement with the experimental data,[18] as can be seen in the figure. According to the present calculation, the eutectoid reaction (βSc) ⇌ (βCr) + (αSc) is occurred at 1093 °C, 65 at.% Sc and the eutectic reaction Liquid ⇌ (βCr) + (βSc) is occurred at 993 °C, 92 at.% Sc. The solubility of Sc in (βCr) and of Cr in (βSc) at 1093 °C is 0.1 and 10 at.%, respectively. The solubility of Sc in (βCr) at 900 °C is 1.2 at.%. The calculated phase diagram fit the experimental data well.

5. Conclusions

The literature data on the Sc-X (X = La, Li, Ca, Cr) binary systems were critically reviewed. The four binary systems Sc-X (X = La, Li, Ca, Cr) were thermodynamically assessed with the experimental information in the literature. Four sets of self-consistent parameters for the four

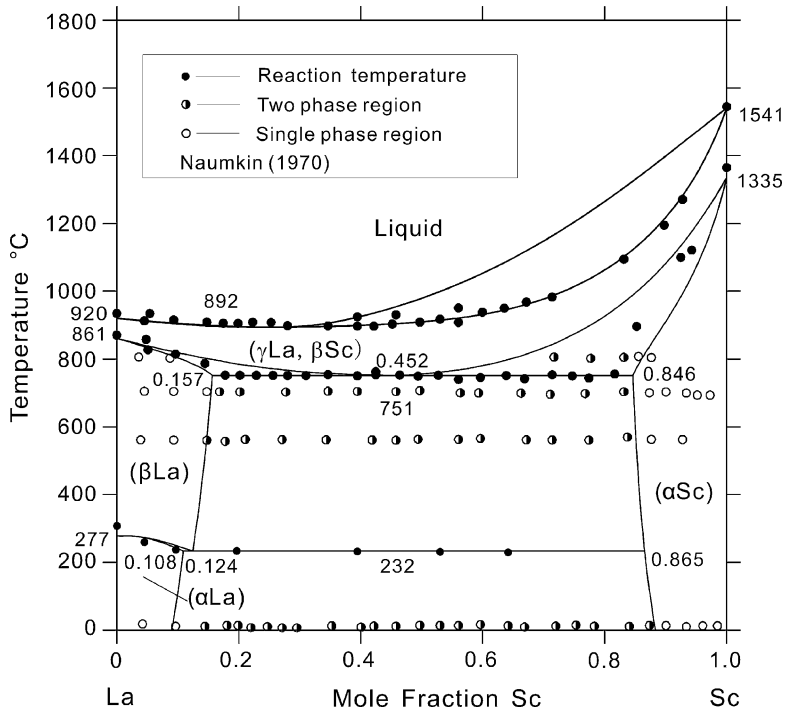

Fig. 1 The calculated phase diagram of the Sc-La system, compared with the experimental data of Ref 11

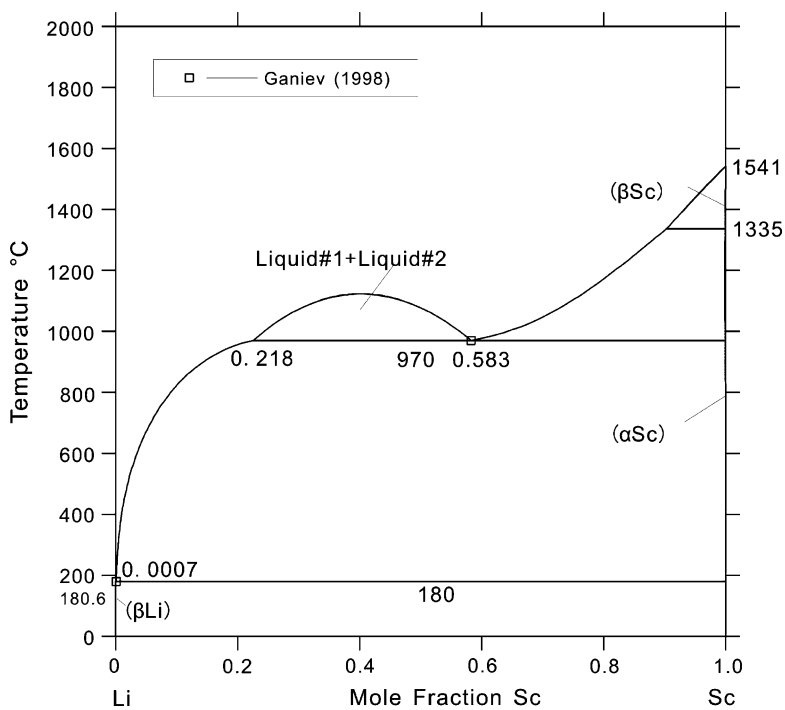

Fig. 2 The calculated phase diagram of the Sc-Li system in comparison with the available experimental data from Ref 15

binary systems were obtained by using the CALPHAD approach. Comparisons between the calculated and measured phase diagrams showed that the experimental data are satisfactorily accounted for by the present thermodynamic description. The presently obtained thermodynamic parameters for the four binary systems are our continuing effort to establish a thermodynamic database of multi-component Al alloys.

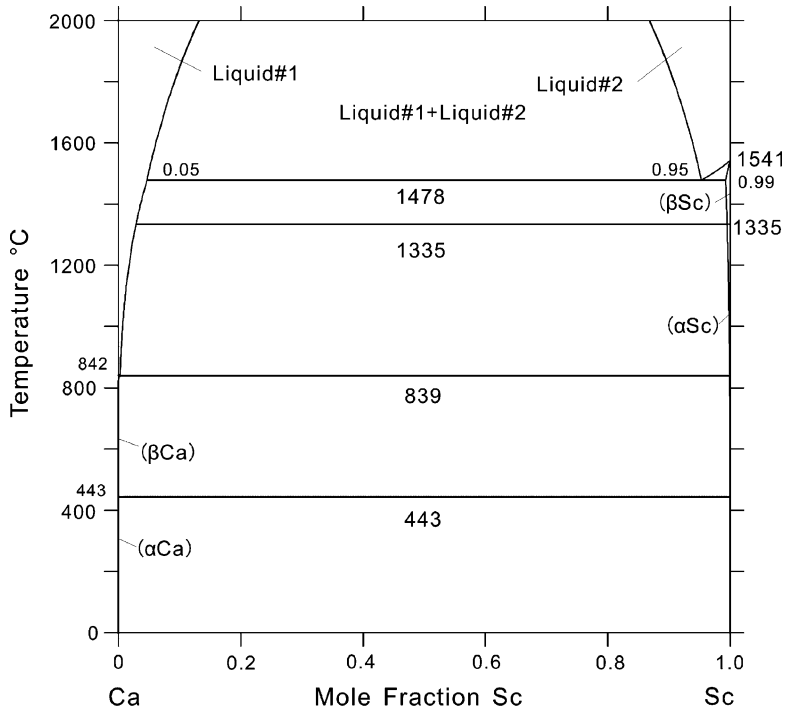

Fig. 3 The calculated phase diagram of the Sc-Ca system

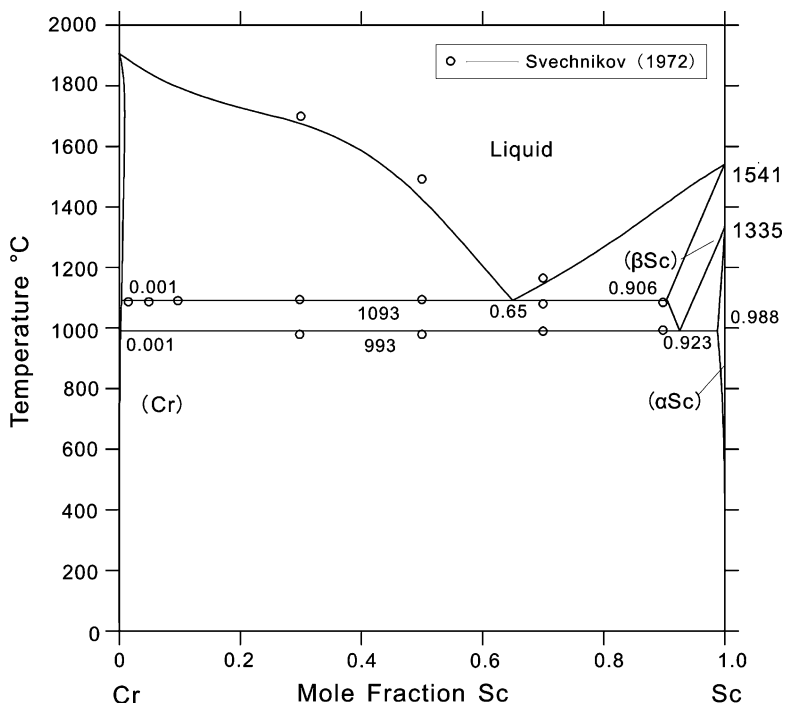

Fig. 4 The calculated phase diagram of the Sc-Cr system, compared with the experimental data of Ref 18

Acknowledgments

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